

Calculator.cs

namespace CalculatorLibrary

{

public class Calculator

{

// Addition

public double Add(double a, double b)

{

return a + b;

}

// Subtraction

public double Subtract(double a, double b)

{

return a - b;

}

// Multiplication

public double Multiply(double a, double b)

{

return a * b;

}

// Division with exception handling

public double Divide(double a, double b)

{

if (b == 0)

{

throw new DivideByZeroException("Cannot divide by zero");

}

```
        return a / b;
    }
}
}
```

Unit Test Class

CalculatorTests.cs

```
using Microsoft.VisualStudio.TestTools.UnitTesting;
using CalculatorLibrary;
using System;
```

```
namespace CalculatorTests
```

```
{
    [TestClass]
    public class CalculatorTests
    {
        private Calculator _calculator;

        [TestInitialize]
        public void Setup()
        {
            _calculator = new Calculator();
        }
    }
}
```

```
//  Addition Tests
```

```
[TestMethod]
public void Add_ShouldReturnCorrectSum()
{
    double result = _calculator.Add(5, 3);
    Assert.AreEqual(8, result);
}
```

```
}
```

```
[TestMethod]
```

```
public void Add_WithZero_ShouldReturnSameNumber()
```

```
{
```

```
    double result = _calculator.Add(5, 0);
```

```
    Assert.AreEqual(5, result);
```

```
}
```

```
//  Subtraction Tests
```

```
[TestMethod]
```

```
public void Subtract_ShouldReturnCorrectDifference()
```

```
{
```

```
    double result = _calculator.Subtract(10, 4);
```

```
    Assert.AreEqual(6, result);
```

```
}
```

```
//  Multiplication Tests
```

```
[TestMethod]
```

```
public void Multiply_ShouldReturnCorrectProduct()
```

```
{
```

```
    double result = _calculator.Multiply(6, 2);
```

```
    Assert.AreEqual(12, result);
```

```
}
```

```
[TestMethod]
```

```
public void Multiply_WithZero_ShouldReturnZero()
```

```
{
```

```
    double result = _calculator.Multiply(6, 0);
```

```
    Assert.AreEqual(0, result);
```

```
}
```

```
//  Division Tests

[TestMethod]
public void Divide_ShouldReturnCorrectQuotient()
{
    double result = _calculator.Divide(10, 2);
    Assert.AreEqual(5, result);
}

[TestMethod]
[ExpectedException(typeof(DivideByZeroException))]
public void Divide_ByZero_ShouldThrowException()
{
    _calculator.Divide(10, 0);
}

//  Edge Case Test

[TestMethod]
public void Subtract_SameNumbers_ShouldReturnZero()
{
    double result = _calculator.Subtract(5, 5);
    Assert.AreEqual(0, result);
}
}
```